inclass-L34

Thursday, October 15, 2026 10:33 AM



inclass-L34

CV2090
Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

Dr. Prakash S Badal

Announcement

- Class/Tutorial rescheduling
  - No Class on 19th Oct
  - No tutorial on 19th Oct
  -
  - 22nd Oct: Tutorial 10-11 am

L-34 (2/; Reg.)
Oct 15, 2026

Earlier we saw

Given (x_i, y_i)

obj: to obtain a 'best' fit line $y' = \alpha + \beta x'$

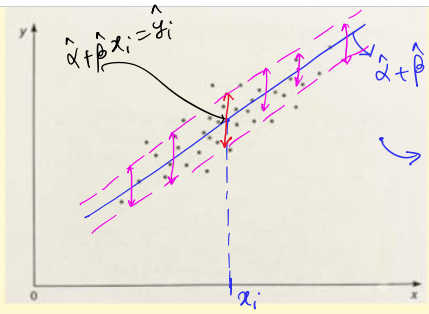
Minimize Ordinary Least-Square error

$$\Delta^2 = \sum (y_i - y_i')^2$$

$\frac{\partial \Delta^2}{\partial \alpha} = 0$ & $\frac{\partial \Delta^2}{\partial \beta} = 0$

Thus, we obtained $\hat{\beta}_{OLS} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$; $\hat{\alpha}_{OLS} = \bar{y} - \hat{\beta}\,\bar{x}$

And the Regression of Y on X becomes $\boxed{\hat{y} = \hat{\alpha} + \hat{\beta} x}$



Conditional variance,

$$s^2_{Y|X=x_i} = \frac{\Delta^2}{n-2} = \frac{\sum (y_i - \hat{y}_i)^2}{n-2}$$

Constant conditional variance assumption

HOMO-SCEDASTICITY

$y_i - \hat{y}_i \rightarrow$ Residual

$\Delta^2 \rightarrow$ Total cum. squared error

$s_{Y|x} \rightarrow$ conditional stdev

CONTRAST IT WITH $s_Y^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$

UNCONDITIONAL VARIANCE

Note on variance

$$s^2_{Y|x} = \frac{\Delta^2}{n-2} = \frac{\sum (Y_i - \hat{Y}_i)^2}{n-2}$$

$$s^2_Y = \frac{\sum (Y_i - \bar{Y})^2}{n-1}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Recall, $\hat{\beta}_{OLS} = \frac{\sum (Y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$

Which one is larger?

$$\sum (Y_i - \hat{Y}_i)^2 = \sum [(Y_i - \bar{y}) - (\hat{y}_i - \bar{y})]^2$$

$$= \sum [(Y_i - \bar{y}) - (\hat{\beta} x_i - \hat{\beta}\bar{x})]^2$$

$$= \sum (Y_i - \bar{y})^2 + \hat{\beta}^2 \sum (x_i - \bar{x})^2 - 2\hat{\beta} \sum (Y_i - \bar{y})(x_i - \bar{x})$$

$$= \sum (Y_i - \bar{y})^2 - \hat{\beta}^2 \sum (x_i - \bar{x})^2$$

$\Rightarrow$ UNEXPLAINED VARIATION IN Y.

TOTAL VARIATION IN Y

EXPLAINABLE VARIATION IN Y } due to Regression

Note on variance

$$\Delta^2 = \sum (Y_i - \bar{y})^2 - \hat{\beta}^2 \sum (x_i - \bar{x})^2$$

Residual $\equiv$ TOTAL VARIATION $-$ EXPLAINED VARIATION

$$s^2_{Y|x} < s^2_Y$$

Confidence interval of regression equation

$$\frac{\bar{Y}_i - \mu_{Y|x_i}}{s_{Y|x}\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}} \sim t_{n-2}$$

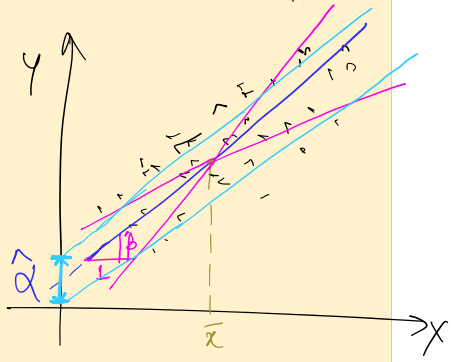
$\mu_{Y|x_i} \rightarrow \hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$

$\frac{1}{n}$ → due to error in $\hat{\alpha}$

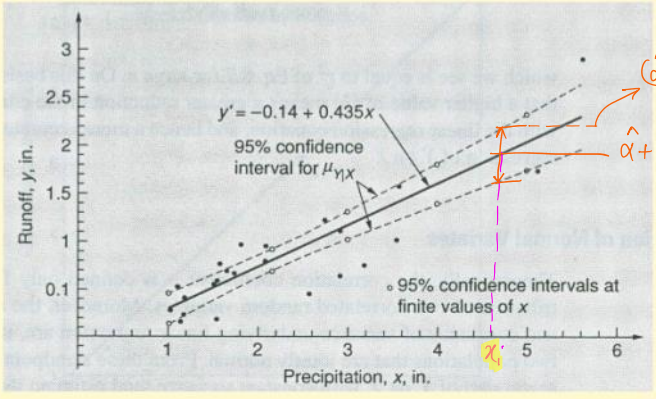
$\frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}$ → due to error in $\hat{\beta}$

$100(1-\alpha)\%$ CONFIDENCE INTERVAL

$$\hat{y}_i \pm t_{1-\frac{\alpha}{2}, n-2}\, s_{Y|x}\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$



Confidence interval of regression equation



$(\hat{\alpha} + \hat{\beta} x_i) \pm t_{1-\frac{\alpha}{2}, n-2}\, s_{Y|x}\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}}$

$\hat{\alpha} + \hat{\beta} x_i$

Statistical Correlation R^2

Ang and Tang (Section 8.2.2, 8.3)

Correlation coefficient

Correlation coefficient

$$\hat{\beta}_{\mathrm{OLS}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \qquad \hat{\alpha}_{\mathrm{OLS}} = \bar{y} - \hat{\beta}_{\mathrm{OLS}}\bar{x}$$

$$\hat{\rho}_{XY} = \frac{1}{n-1}\frac{\sum x_i y_i - n\bar{x}\bar{y}}{s_X s_Y}$$

Storm No.	Precip., x_i (in.)	Runoff, y_i (in.)	$x_i y_i$	x_i^2	y_i^2	$y_i' = \hat{\alpha} + \hat{\beta}x_i$	$(y_i - y_i')^2$
1	1.11	0.52	0.58	1.23	0.270	0.343	0.0313
2	1.17	0.40	0.47	1.37	0.160	0.369	0.0009
3	1.79	0.97	1.74	3.20	0.941	0.637	0.1110
4	5.62	2.92	16.40	31.60	8.526	2.280	0.4000
5	1.13	0.17	0.19	1.28	0.029	0.351	0.0328
6	1.54	0.19	0.29	2.37	0.036	0.530	0.1158
7	3.19	0.76	2.43	10.15	0.578	1.245	0.2360
8	1.73	0.66	1.14	2.99	0.436	0.612	0.0023
9	2.09	0.78	1.63	4.37	0.608	0.770	0.0001
10	2.75	1.24	3.41	7.55	1.538	1.059	0.0328
11	1.20	0.39	0.47	1.44	0.152	0.381	0.0001
12	1.01	0.30	0.30	1.02	0.090	0.299	0.0000
13	1.64	0.70	1.15	2.69	0.490	0.574	0.0158
14	1.57	0.77	1.21	2.46	0.593	0.544	0.0511
15	1.54	0.59	0.91	2.37	0.348	0.530	0.0036
16	2.09	0.95	1.99	4.36	0.902	0.770	0.0326
17	3.54	1.02	3.62	12.55	1.040	1.400	0.1442
18	1.17	0.39	0.46	1.37	0.152	0.368	0.0004
19	1.15	0.23	0.26	1.32	0.053	0.360	0.0169
20	2.57	0.45	1.16	6.60	0.202	0.980	0.2810
21	3.57	1.59	5.66	12.74	2.528	1.415	0.0306
22	5.11	1.74	8.90	26.18	3.028	2.084	0.1185
23	1.52	0.56	0.85	2.31	0.314	0.521	0.0015
24	2.93	1.12	3.28	8.58	1.254	1.135	0.0002
25	2.93	0.64	0.74	1.34	0.410	0.365	0.0755
	$\sum = 53.89$	20.05	59.24	153.44	24.678		$\Delta^2 = 1.7350$

$$\bar{x} = \frac{53.89}{25} = 2.16; \quad \bar{y} = \frac{20.05}{25} = 0.80$$

$$s_X^2 = \frac{1}{24}(153.44 - 25 \times 2.16^2) = 1.533;$$

$$s_Y^2 = \frac{1}{24}(24.68 - 25 \times 0.80^2) = 0.362$$

$$\hat{\beta} = \frac{59.24 - 25 \times 2.16 \times 0.80}{153.44 - 25(2.16)^2} = 0.435;$$

$$\hat{\alpha} = 0.80 - 0.435 \times 2.16 = -0.140$$

$$s_{Y|X}^2 = \frac{\Delta^2}{n-2} = \frac{1.735}{25-2} = 0.075$$

$$\hat{\rho} = \frac{1}{24}\frac{59.24 - 25 \times 2.16 \times 0.80}{\sqrt{1.533}\sqrt{0.362}} = 0.898$$

We can also establish the 95% confidence interval as follows: For this purpose, let us select five discrete values of the precipitation at $x_i = 1.0$, 2.16, 3.0, 4.0, and 5.0 in. Then, according to Eq. 8.8, the respective confidence intervals at these five different values of x_i are, individually, with $t_{0.975,23} = 2.069$ from Table A.3.

$$\text{At } x_i = 1.0: \quad \langle \mu_{Y|1.0} \rangle_{0.95} = 0.295 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(1.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (0.138 \rightarrow 0.452) \text{ in.}$$

$$\text{At } x_i = 2.16: \quad \langle \mu_{Y|2.16} \rangle_{0.95} = 0.800 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(2.16 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (0.687 \rightarrow 0.913) \text{ in.}$$

$$\text{At } x_i = 3.0: \quad \langle \mu_{Y|3.0} \rangle_{0.95} = 1.165 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(3.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (1.027 \rightarrow 1.303) \text{ in.}$$

$$\text{At } x_i = 4.0: \quad \langle \mu_{Y|4.0} \rangle_{0.95} = 1.600 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(4.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (1.368 \rightarrow 1.832) \text{ in.}$$

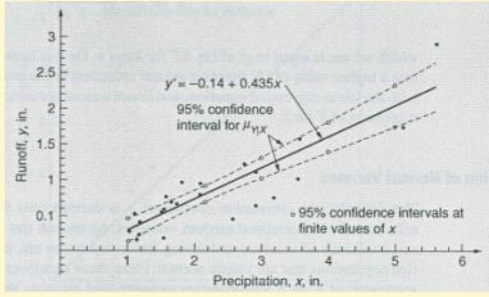
$$\text{At } x_i = 5.0: \quad \langle \mu_{Y|5.0} \rangle_{0.95} = 2.035 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(5.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (1.746 \rightarrow 2.324) \text{ in.}$$

$$E(Y|X = 4) = -0.14 + 0.435 \times 4 = 1.6 \text{ in.}$$

$$P(Y > 2|X = 4) = 1 - \Phi\left(\frac{2 - 1.6}{0.274}\right) = 1 - 0.928 = 0.072$$

$100(1 - \alpha)\%$ CI for the regression at $X = x_i$:

$$\bar{y}_i \pm t_{(1-\alpha/2),n-2} s_{Y|x} \sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$



Linear Reg. with Non-constant σ

Heteroscedasticity

Ang and Tang (Section 8.4)

Thank you!